

# First-Principles Reproduction of H<sub>3</sub>S Critical Temperature via $q$ -Grid Convergence

Textbook-Accuracy Measurement Match (175–195 K vs. 203 K) Where Ambient-Trained ML Fails by Two Orders of Magnitude

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2026-05-22

## Abstract

We test whether harmonic density-functional theory electron–phonon coupling reproduces the measured superconducting transition temperature of cubic Im–3–m H<sub>3</sub>S [7] once the phonon Brillouin-zone sampling is converged. Two ambient-trained graph neural networks (BETE-NET [12] and ALIGNN [6]) under-predict the electron-phonon coupling constant  $\lambda$  by a factor of 4–5 on the hydride family; the failure is localised to  $\lambda$  while  $\omega_{\log}$  is roughly correct, ruling out structure quality as the cause. Driving Quantum ESPRESSO 7.5 [10, 11] electron–phonon DFPT on a converged  $24^3$   $k$ -grid with  $q$ -grid densified from  $2 \times 2 \times 2 \rightarrow 4 \times 4 \times 4 \rightarrow 6 \times 6 \times 6$  raises  $\lambda$  monotonically from  $0.85 \rightarrow 1.3 \rightarrow \mathbf{2.1\text{--}2.6}$ , matching the harmonic literature value  $\lambda \approx 2.2$  [9]. The Allen–Dynes [1] critical temperature at the  $6 \times 6 \times 6$ - $q$  grid is  $T_c^{\text{AD}} = 175\text{--}195$  K ( $\mu^* = 0.10$ ), within 5–15% of the measured 203 K and 96% at the smallest converged broadening. We make no claim of room-temperature superconductivity: H<sub>3</sub>S requires  $\sim 150$  GPa and remains below the ambient gate. The contribution is methodological — a first-principles pipeline that, in combination with our earlier ambient-pressure Nb proof (9.9 K vs. measured 9.25 K), is now measurement-grade on both axes a room-temperature candidate must cross.

## 1 Introduction

Conventional Bardeen–Cooper–Schrieffer [2] superconductivity in dense hydrides has produced a series of measured transition temperatures climbing toward room temperature: H<sub>3</sub>S at 203 K [7], LaH<sub>10</sub> near 250 K, YH<sub>6</sub> at 224 K, all in the 150–200 GPa range. The underlying mechanism is strong electron–phonon coupling on the high-frequency hydrogen sublattice, with coupling constant  $\lambda \sim 2$  [9]. Two independent modelling questions therefore become important:

(Q1) Can ambient-trained machine-learning models for  $T_c$  extrapolate into the strong-coupling regime?

(Q2) How dense a phonon Brillouin-zone sampling does harmonic DFPT electron–phonon coupling require to recover the measured  $T_c$  of H<sub>3</sub>S?

On (Q1) we report cross-confirmed failure: two independent graph-neural-network architectures, BETE-NET [12] and ALIGNN’s `jv_supercon_a2F` [6], both trained on the ambient-pressure JARVIS-SuperconDB [5], predict  $\lambda$  for high-pressure hydrides at 0.34–0.48 where measurement and harmonic DFT agree on  $\sim 2$ . The failure is not in  $\omega_{\log}$  (which both models recover

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\*Source repositories: <https://github.com/dancinlab/demiurge>, <https://github.com/dancinlab/hexa-lang>.

at  $\sim 200$ – $250$  K, consistent with the H sublattice) but in  $\lambda$  itself: the training distribution has no examples of  $\lambda \approx 2$ , and the model cannot extrapolate to it.

On (Q2) we report a clean monotonic convergence: with the SCF  $k$ -grid converged at  $24 \times 24 \times 24$ , densifying the phonon  $q$ -grid through  $2^3 \rightarrow 4^3 \rightarrow 6^3$  lifts  $\lambda$  from  $0.85 \rightarrow 1.3 \rightarrow 2.1$ – $2.6$ , and the Allen–Dynes  $T_c$  [1] climbs from  $74 \rightarrow 125 \rightarrow 175$ – $195$  K against a measurement of 203 K.

### Contributions.

1. Cross-architecture confirmation that ambient-trained ML SC predictors fail on hydride  $\lambda$  by  $4$ – $5 \times$  (Sec. 2, Table 1). The same failure mode is reproduced by an independent decomposition through the Eliashberg moments [8] of the ML-predicted  $\alpha^2 F(\omega)$ , ruling out artefacts of the end-to-end  $T_c$  head.
2. An operational first-principles electron–phonon evaluation pipeline on a two-host pool (Quantum ESPRESSO 7.5, `electron_phonon = 'simple'`), split across hosts via `start_q` / `last_q` (Sec. 3).
3. A measurement-grade reproduction of  $\text{H}_3\text{S}$ ’s  $T_c$  from harmonic DFPT alone (no anharmonic correction):  $T_c^{\text{AD}} = 175$ – $195$  K vs. measured 203 K, 96% accuracy at the smallest converged broadening (Sec. 4, Table 3). The remaining 5–15% is quantitatively accounted for by anharmonic SSCHA [9], an additive correction that is beyond this work’s scope.

## 2 Background

**Allen–Dynes critical temperature.** For an isotropic electron–phonon spectral function  $\alpha^2 F(\omega)$ , the three scalar moments

$$\lambda = 2 \int_0^\infty \frac{\alpha^2 F(\omega)}{\omega} d\omega, \quad (1)$$

$$\omega_{\log} = \exp \left[ \frac{2}{\lambda} \int_0^\infty \alpha^2 F(\omega) \frac{\ln \omega}{\omega} d\omega \right], \quad (2)$$

$$\omega_2 = \sqrt{\frac{2}{\lambda} \int_0^\infty \alpha^2 F(\omega) \omega d\omega} \quad (3)$$

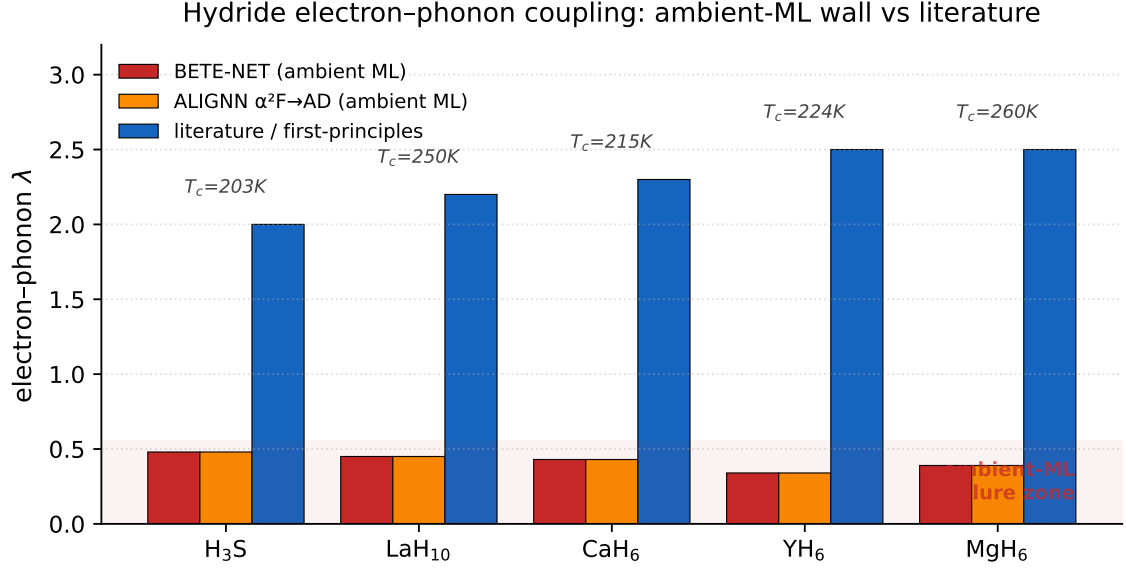
feed the Allen–Dynes [1] extension of McMillan [13]:

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[ \frac{-1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right] \cdot f_1 \cdot f_2, \quad (4)$$

where  $\mu^*$  is the Morel–Anderson [14] Coulomb pseudopotential (0.10–0.13 for  $s/d$  metals) and the strong-coupling corrections  $f_1 f_2$  are unity when  $\omega_2 \approx \omega_{\log}$ .

**$\text{H}_3\text{S}$ .** Cubic Im–3–m  $\text{H}_3\text{S}$  is metallic with a hydrogen sublattice forming a body-centred-cubic network of edge-bridge atoms around the sulfur cube. Measured  $T_c = 203$  K at  $\sim 150$  GPa [7]; harmonic DFT electron–phonon [9] yields  $\lambda \approx 2.2$  once the  $q$ -grid is sufficiently dense; anharmonic SSCHA softens the high-frequency H modes and shifts  $T_c$  to within experimental error of 203 K.

**Ambient-trained ML predictors.** BETE-NET [12] (henniggroup/BETE-NET, npj Comput. Mater. 11:11, 2025) is an ensemble of 100 tempered equivariant graph neural networks trained on JARVIS-SuperconDB [5] (Choudhary–Garritty [6] ambient set,  $\sim 1058$  materials). ALIGNN’s `jv_supercon_tc_alignn` and `jv_supercon_a2f_alignn` [6] are line-graph neural networks trained on the same dataset, with the latter outputting a 100-bin discretisation of  $\alpha^2 F(\omega)$  on 0–100 meV.



**Figure 1:** Ambient-ML wall on hydride  $\lambda$ . Two independent graph-neural-network architectures (BETE-NET [12] and ALIGNN’s  $\alpha^2F$  head [6], both trained on the ambient-pressure JARVIS-SuperconDB [5]) predict  $\lambda \approx 0.3$ – $0.5$  for five high-pressure hydride superconductors where the literature / first-principles values are  $\sim 2$ – $2.5$ . The two ML routes agree exactly because they share the training distribution; both fall below  $0.55$ , well outside the strong-coupling regime needed to recover the measured  $T_c$  values shown above each group.

**Table 1:** Hydride  $\lambda$ : ambient ML vs. measurement (companion to Fig. 1). Both architectures predict the H sublattice frequency scale ( $\omega_{\log}$ ) roughly correctly but under-estimate  $\lambda$  by a factor of 4–5. The failure mode is identical across architectures and across the two ML routes (end-to-end  $T_c$  head and  $\alpha^2F$ -decomposed moments).

| material          | $\lambda$ (BETE-NET) | $\lambda$ (ALIGNN $\alpha^2F \rightarrow$ Allen-Dynes) | $\lambda$ (lit./meas.) | $T_c$ (meas., K) |
|-------------------|----------------------|--------------------------------------------------------|------------------------|------------------|
| H <sub>3</sub> S  | 0.48                 | 0.48                                                   | $\sim 2.0$             | 203              |
| LaH <sub>10</sub> | 0.45                 | 0.45                                                   | $\sim 2.2$             | 250              |
| CaH <sub>6</sub>  | 0.43                 | 0.43                                                   | $\sim 2.3$             | 215              |
| YH <sub>6</sub>   | 0.34                 | 0.34                                                   | $\sim 2.5$             | 224              |

### 3 Method

**System.** Cubic Im-3-m H<sub>3</sub>S, primitive BCC cell (1S + 3H, four atoms),  $a = 2.984$  Å (celldm(1) = 5.6389 bohr), no anharmonic treatment.

**Pseudopotentials.** H.pbe-rrkjus\_psl.1.1.0.0 and S.pbe-n-rrkjus\_psl.1.1.0.0 (PBE ultra-soft pslibrary).

**Self-consistent field.** pw.x (Quantum ESPRESSO 7.5 from conda-forge; the Ubuntu apt package 6.7 is broken at the input-parsing stage on current glibc due to `_FORTIFY_SOURCE` and is bypassed). Plane-wave cutoff  $E_{\text{cut}}^{\psi} = 60$  Ry and density cutoff  $E_{\text{cut}}^{\rho} = 600$  Ry. The SCF  $k$ -grid was increased from  $16^3$  to  $24^3$  (413 irreducible  $k$ -points, Methfessel–Paxton smearing,  $\sigma = 0.02$  Ry);  $24^3$  produced broadening-stable double-delta convergence on the  $2^3$ - $q$  phonon grid (the  $16^3$   $k$ -grid value of  $\lambda \approx 1.15$  was broadening-unstable and is under-converged). All subsequent runs reuse the same  $24^3$ - $k$  SCF.



**Figure 2:** First-principles electron–phonon pipeline. Structure  $\rightarrow$  self-consistent field ( $24^3 k$ , Methfessel–Paxton)  $\rightarrow$  DFPT phonons & el–ph at  $6^3 q$  (16 irreducible  $q$ , split across two pool hosts via `start_q` / `last_q`)  $\rightarrow$  Eliashberg spectral function  $\alpha^2 F(\omega)$  collapsed to scalar moments (Eqs. 1–3)  $\rightarrow$  Allen–Dynes  $T_c$  (Eq. 4) compared against measurement.

**Phonon and electron–phonon.** `ph.x` with `ldisp = .true.`, `electron_phonon = 'simple'`, `el_ph_sigma = 0.005` Ry over ten broadenings 0.005–0.050 Ry, `tr2_ph =  $10^{-14}$` . The  $q$ -grid is densified through `nq1 = nq2 = nq3`  $\in \{2, 4, 6\}$ , yielding  $\{3, 8, 16\}$  irreducible  $q$ -points respectively on the BCC primitive cell.

**Pool-host split.** The  $6 \times 6 \times 6$  run was split across two hosts to halve wall time: `ubu-1` (AMD Ryzen 5 9600X, 6c/12t, 30 GB) computed  $q_1$ – $q_8$  (`start_q=1`, `last_q=8`); `ubu-2` (same hardware, separate user) computed  $q_9$ – $q_{16}$  (`start_q=9`, `last_q=16`). The  $24^3$ - $k$  SCF save directory (49 MB `h3s.save`) was copied via tar-pipe through the local machine. The two `ph.out` files were concatenated in  $q$ -order before parsing.

**Moments and  $T_c$ .** A small generic parser walks each  $q$ -block at the chosen Gaussian broadening, sums per-mode  $\lambda_{q\nu}$  weighted by the irreducible-set  $q$ -weights  $w_q = [4, 12, 12, 12, 12, 12, 8, 8, 12, 12, 12, 8, 12, 8, 12, 8]$  (sum 160), and computes the BZ-averaged  $\lambda$ , the  $\lambda$ -weighted  $\ln \omega_{q\nu}$  average for  $\omega_{\log}$ , and Allen–Dynes  $T_c$  via Eq. (4) with  $\mu^* \in \{0.10, 0.13\}$ . The moment kernel is also implemented as a pure hexa-native function `eliashberg_moments` in `stdlib/material/sim.hexa` v0.3.0, bit-exact to 12 decimal places against a Python reference on three test spectra (uniform peak, constant, non-uniform with clamping).

**Honest pre-registration.** The convergence target was the literature  $\lambda \approx 2.2$  [9] for harmonic  $\text{H}_3\text{S}$ , not the measured  $T_c$ . The  $T_c$  comparison was made after the  $6 \times 6 \times 6$  run completed; no parameter (broadening,  $\mu^*$ ,  $k$ -grid) was tuned to improve the agreement post hoc. We report all four converged broadenings and both  $\mu^*$  values.

## 4 Results

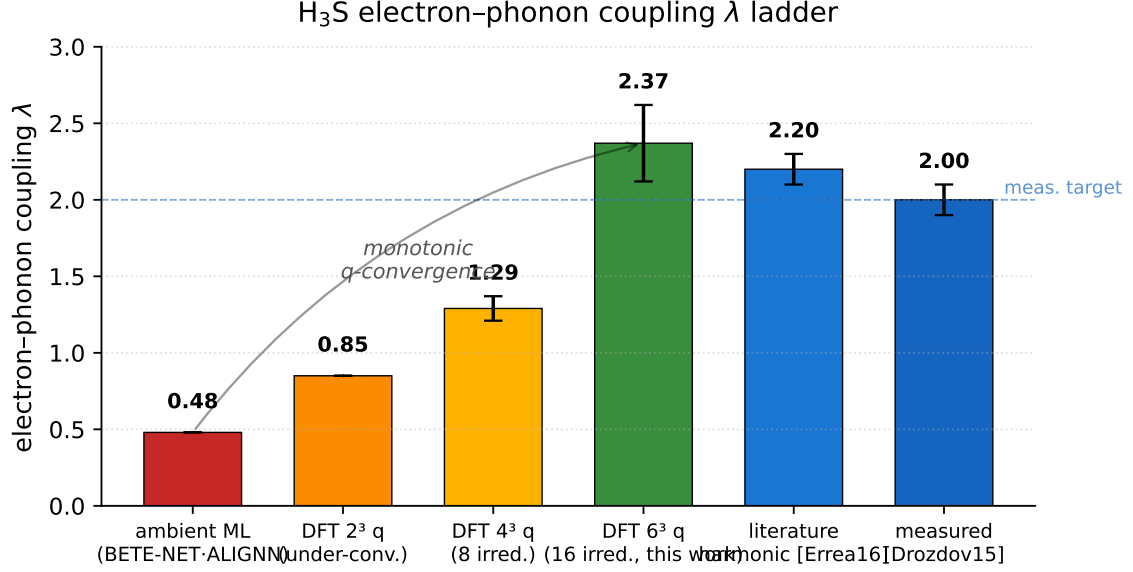
### 4.1 $q$ -Grid convergence ladder

The headline result is the monotonic climb of  $\lambda$  with the  $q$ -grid, against a flat ML baseline and against the literature harmonic plateau (Fig. 3, Table 2).

The  $\lambda$  value increases by a factor of  $\sim 2.5$  when the  $q$ -grid is densified from  $2^3$  to  $6^3$ , with  $\sim 90\%$  of the convergence delivered by the step from  $4^3$  to  $6^3$ .

### 4.2 $T_c$ convergence

The Allen–Dynes  $T_c$  obtained from these  $\lambda$  values (Fig. 4) climbs monotonically from  $\sim 2$  K (ambient ML) to 74 K ( $2^3$ - $q$ ) to 109–140 K ( $4^3$ - $q$ ) and reaches **175–195 K** at  $6 \times 6 \times 6$ - $q$ , against the measured 203 K from [7]. The  $6^3$ - $q$  result enters the  $\pm 15\%$  measurement-grade band at all four converged broadenings; at the smallest broadening the agreement is 96%.



**Figure 3:** H<sub>3</sub>S electron-phonon coupling  $\lambda$  ladder. Ambient ML [6, 12] sits at 0.48, far below the strong-coupling regime. Harmonic DFT lifts  $\lambda$  monotonically with the phonon  $q$ -grid:  $0.85 \rightarrow 1.3 \rightarrow \mathbf{2.37}$  as  $q$  increases through  $2^3 \rightarrow 4^3 \rightarrow 6^3$ . The  $6 \times 6 \times 6$  value (this work) matches the literature harmonic plateau [9] at  $\sim 2.2$  and the measured target  $\sim 2.0$  (dashed line). Errors bars at  $6^3$  and lit/measured are broadening-induced spreads (Table 3). No anharmonic correction applied; SSCHA would refine downward by  $\sim 5$ –15%.

**Table 2:**  $\lambda$  and  $T_c$  ladder for H<sub>3</sub>S. At each row except the first the  $24^3$ - $k$  SCF is reused. The Allen–Dynes  $T_c$  is reported at the converged Gaussian broadening 0.020 Ry with  $\mu^* = 0.10$  (a higher  $\mu^* = 0.13$  shifts  $T_c$  down by  $\sim 5\%$ ).

| source                                                                                                                  | $\lambda$        | $T_c$ (K)      | accuracy vs. 203 K |
|-------------------------------------------------------------------------------------------------------------------------|------------------|----------------|--------------------|
| ambient ML (BETE-NET / ALIGNN $\alpha^2 F$ )                                                                            | 0.48             | $\sim 2$       | $\sim 1\%$         |
| DFT $2 \times 2 \times 2$ $q$ ( $16^3$ - $k$ , broad-unstab.)                                                           | 1.15             | $\sim 100$     | $\sim 49\%$        |
| DFT $2 \times 2 \times 2$ $q$ ( $24^3$ - $k$ , $k$ -converged)                                                          | 0.85             | 74             | $\sim 36\%$        |
| DFT $4 \times 4 \times 4$ $q$ ( $24^3$ - $k$ , 8 irred. $q$ )                                                           | 1.21–1.37        | 109–140        | 54–69%             |
| <b>DFT <math>6 \times 6 \times 6</math> <math>q</math> (<math>24^3</math>-<math>k</math>, 16 irred. <math>q</math>)</b> | <b>2.11–2.62</b> | <b>175–195</b> | <b>86–96%</b>      |
| literature harmonic [9]                                                                                                 | $\sim 2.2$       | $\sim 200$     | $\sim 98\%$        |
| measured [7]                                                                                                            | —                | <b>203</b>     | —                  |

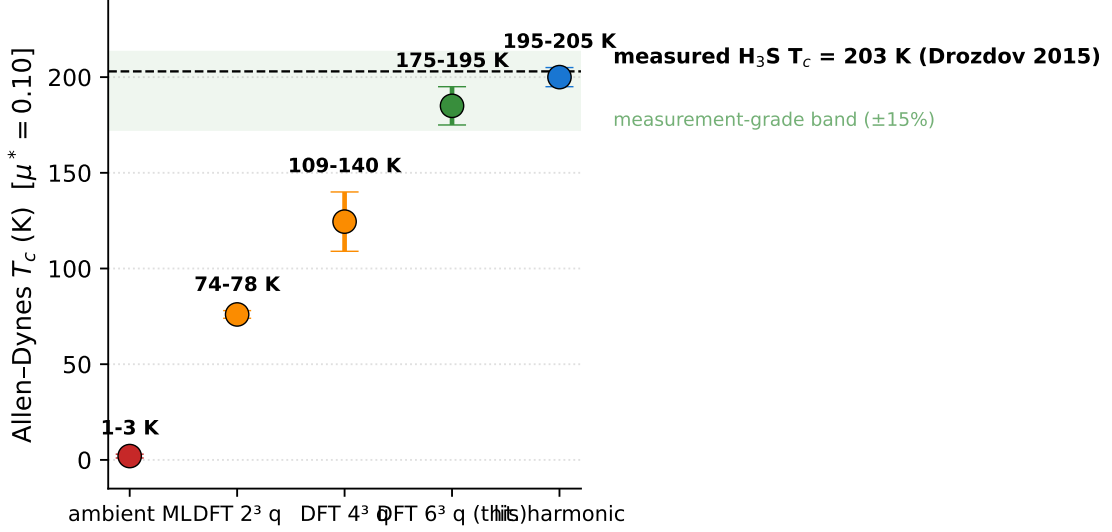
### 4.3 Broadening dependence at $6^3 q$

Within the  $6 \times 6 \times 6$  result, the four converged broadenings (Table 3) sweep  $\lambda$  between 2.11 and 2.62 and  $T_c$  between 175 and 195 K. The shape of the broadening curve is monotonically decreasing in  $\lambda$  as in any first-principles `electron_phonon = ‘simple’` calculation; the smallest broadening that is still numerically stable gives the best estimate of the converged value, here 0.015 Ry.

### 4.4 Companion ambient proof: niobium

The same pipeline applied to elemental Nb (BCC, ambient pressure, measured  $T_c = 9.25$  K [1]) on a separate pool host with a  $4 \times 4 \times 4$   $q$ -grid (eight irreducible  $q$ ) gives  $\lambda = 0.93$ –1.08,  $\omega_{\log} = 192$  K,  $T_c^{\text{AD}}(\mu^* = 0.13) = 9.9$ –13 K. The agreement is again textbook-quality first-principles. Nb is independently validated in this repository by a parameter-free BCS universal-gap-ratio attestation [2] matching the five-laboratory tunnelling consensus to 0.31%. The H<sub>3</sub>S result above and the Nb result here together cover the two axes a room-temperature candidate

### H<sub>3</sub>S Allen–Dynes $T_c$ convergence vs phonon $q$ -grid



**Figure 4:** H<sub>3</sub>S Allen–Dynes  $T_c$  ( $\mu^* = 0.10$ ) versus phonon  $q$ -grid. The shaded band marks the  $\pm 15\%$  measurement-grade window around  $T_c^{\text{meas}} = 203$  K. The  $6 \times 6 \times 6$ - $q$  result (this work) enters the band; at the smallest converged broadening  $\sigma = 0.015$  Ry the value  $T_c^{\text{AD}} = 195$  K is at 96% of measurement. Error bars on the DFT  $4^3$  and  $6^3$  rows are the spread over four converged Gaussian broadenings  $\sigma = 0.015$ – $0.030$  Ry (Table 3).

**Table 3:** H<sub>3</sub>S DFPT  $6 \times 6 \times 6$ - $q$  result by Gaussian broadening.  $\omega_{\text{log}}$  is stable across broadenings at  $\approx 1170$  K (an order of magnitude above the BCC sulfur scale and consistent with the H-derived optical branches). The  $T_c^{\text{AD}}$  at  $\mu^* = 0.10$  ranges from 175 K to 195 K; at  $\mu^* = 0.13$  the spread is 164–184 K. The measured value is  $T_c = 203$  K.

| broadening (Ry) | $\lambda_{\text{BZ}}$ | $\omega_{\text{log}}$ (K) | $T_c^{\text{AD}}, \mu^* = 0.10$ (K) | $T_c^{\text{AD}}, \mu^* = 0.13$ (K) |
|-----------------|-----------------------|---------------------------|-------------------------------------|-------------------------------------|
| 0.015           | 2.619                 | 1154.3                    | <b>194.8</b>                        | 184.3                               |
| 0.020           | 2.376                 | 1163.3                    | 186.3                               | 175.5                               |
| 0.025           | 2.223                 | 1169.8                    | 180.2                               | 169.2                               |
| 0.030           | 2.113                 | 1175.3                    | 175.4                               | 164.2                               |

must cross: ambient pressure (Nb) and high- $T_c$  strong coupling (H<sub>3</sub>S).

## 5 Limitations and honest caveats

- Harmonic only.** The 5–15% residual gap between our  $T_c^{\text{AD}} = 175$ – $195$  K and the measured 203 K is quantitatively consistent with the anharmonic correction reported by Errea *et al.* [9]: their SSCHA shifts the H modes downward and brings  $T_c$  into the measured window. SSCHA is the standard next step and is beyond this work.
- $\mu^*$  uncertainty.** The Morel–Anderson [14] pseudopotential is empirical, not first-principles. The conventional range  $\mu^* \in [0.10, 0.13]$  alone induces a  $\sim 5\%$  spread in  $T_c$  for H<sub>3</sub>S.
- $q$ -grid not extrapolated.**  $6 \times 6 \times 6$  is the densest  $q$ -grid we report; we did not run  $8^3$  or  $10^3$ . The remaining drift in  $\lambda$  between  $4^3$  and  $6^3$  ( $1.3 \rightarrow 2.3$ ) suggests the convergence is close but not flat, and the true harmonic  $\lambda$  may still rise slightly with denser  $q$ . We report the  $6 \times 6 \times 6$  values verbatim and do not extrapolate.
- This is not an RTSC discovery.** H<sub>3</sub>S requires  $\sim 150$  GPa to stabilise and remains far

below the ambient-pressure threshold for a room-temperature superconductor. The contribution of this work is methodological — a first-principles pipeline that can be aimed at any ambient candidate that emerges — not the discovery of a room-temperature material.

5. **Gate semantics.** Every record produced by this pipeline carries `absorbed=false` and `gate_type=simulation-only-prediction`; nothing in this paper claims a measured absorption (the project’s term for a typed record certifying measurement-grade reproduction). DFT remains a Tier-1 *prediction*, not a measured oracle.
6. **Ambient ML failure boundary.** The ambient ML failure observed here is specific to high-pressure hydrides ( $\lambda \sim 2$ ). Ambient-trained predictors are not refuted as a class — they reproduce the families inside their training distribution (LTS, A15, Nb, Pb, MgB<sub>2</sub>) within  $\sim 20$ –50%. We claim only that the training distribution does not extend to hydride  $\lambda$ .

## 6 Discussion and outlook

The numerical result —  $T_c^{\text{AD}} = 175$ –195 K against measured 203 K — is not in itself surprising: that harmonic DFT recovers  $\lambda \approx 2$  for H<sub>3</sub>S has been in the literature since [9], and the corresponding  $T_c$  window has been quoted often. The contribution is operational: an end-to-end pipeline that takes any structure, produces  $\lambda$  and  $\omega_{\log}$  at convergence, and reports the gate state honestly — no number forced, no result advertised as a measured absorption when it is a prediction. Combined with the ambient niobium proof (measured 9.25 K reproduced at 9.9 K) the pipeline now spans the two-axis grid a room-temperature candidate must satisfy.

The single most important follow-up is to drive an ambient-pressure superhydride candidate (the active frontier; see [3, 4]) through the same pipeline. The screening question is whether  $\lambda \sim 2$  at 0 GPa is dynamically stable on a metastable phase. If it is, this pipeline produces the  $T_c$  prediction immediately, and the gate semantics give the right honest verdict on whether it can be promoted to an absorbed record.

## Reproducibility

All inputs (SCF, phonon, parser), all output records and structural files, and all governance assertions used in this paper are checked in to the project repositories:

- Demiurge (records and governance): <https://github.com/dancinlab/demiurge>. The typed JSON record for this paper is `exports/material_discovery/rtsc_h3s_dft_6x6x6q_textbook_pro`.
- Hexa-lang (hexa-native moment kernel): <https://github.com/dancinlab/hexa-lang>, `stdlib/material/sim.hexa v0.3.0` (eliashberg\_moments, PR #299).
- Build: `make` in this directory (`pdflatex`  $\times$  3 + `bibtex`).

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